Neural DNO meeting notes

October 1, 2026

Main message. Inference is now fast enough to compete with low-order Craig–Sulem evaluation, but we do not yet have a reliable accuracy advantage. The phase-focused recipes improved Stokes substantially and hurt Tanaka. Our next tests target depth conditioning and incomplete supervision of the operator.

Where we stand

Median terminal surface error in percent, 32 test cases per family, against M6. All three neural models below were trained for 120 epochs.

	M2	M3	Original EMA	Local tangent	Modal phase
Stokes	0.000470	0.000144	0.01055	0.002296	0.002385
Tanaka	0.1893	0.01237	0.1241	0.2150	0.1759
Benjamin–Feir	0.4026	0.2267	0.3127	0.3121	0.2923
JONSWAP/TMA	0.01917	0.005882	0.1516	0.1249	0.1019

Accuracy. The new recipes improve Stokes by about 4.5× versus the original EMA model, but neither beats M3's median in any family. Modal phase improves three medians versus local tangent while worsening all four 95th-percentile errors. Tanaka's 95th percentile rises from 4.64% (original) to 7.67% (local) and 11.19% (modal). Both new recipes also changed the data normalization; the Stokes improvement cannot be attributed to phase loss alone.

Runtime. On Ada, the original optimized EMA model takes 1.25–1.29 seconds for $T = 20$ and 12.33–12.62 seconds for $T = 200$. M2 takes approximately 0.97/9.6 seconds and M3 1.57/15.7 seconds. The model sits between them in cost, but generally not in accuracy. These are warm, full single rollouts, not batch-throughput estimates.

What the diagnostics suggest

Phase drift matters, but its loss is not a complete fix. Earlier alignment diagnostics attributed most Stokes/Tanaka trajectory error to translation. The completed experiments show that targeting phase can improve one regime while making steep Tanaka tails worse.

A stronger lead: learning the operator beyond the observed potential. At fixed (η, h), changing only the phases of ξ while preserving its Fourier amplitudes raises the original model's paired median one-step error by 23× on Stokes and 119× on Benjamin–Feir. All three models beat M2 on all 32 original BF inputs but on none of the 32 scrambled-input case medians. This suggests inadequate coverage of Dirichlet directions; it does not yet establish the cause of rollout drift.

Depth conditioning remains plausible. Adding η/h improved all four fresh one-epoch median rollout errors, but worsened Tanaka's 95th percentile. Gauge centering made little difference; learned surface filters were mixed. Positivity held on the four audited hard Tanaka surfaces, so positivity enforcement is not our first intervention.

Running now and next decision

As of 03:22 local time, the η/h candidate has completed 7 of 120 epochs; its otherwise identical target-loss control is queued on GPU 1. Rollouts are scheduled at epochs 10, 30, 60, and 120.

The separate fresh one-epoch pilot tests **CS-labelled potential probes**: keep each surface/depth, retain the original potential, and add a same-spectrum phase-scrambled potential with its own M6 label. Its paired control is running; the probe arm follows. Both use LR 2×10^{-5} and receive 128-case rollout evaluation. This changes training coverage, not inference cost.

Decision criterion. Compare the controlled runs using family medians *and* tails. Look for gains that survive rollout evaluation without a Tanaka regression before committing to another long run. Do not promote a variant solely because its aggregate validation loss is lower.

Figure 12: each plotted accuracy is one selected test case, not the 32-case medians above. Orange points pair measured runtime and accuracy for the epoch-120 EMA checkpoint; A100 and Blackwell were retimed on October 1. Sources: September 30 and October 1 experiment logs, `outputs/phase_timing_20260930/`, and `outputs/operator_probe_a100_20261001/`.